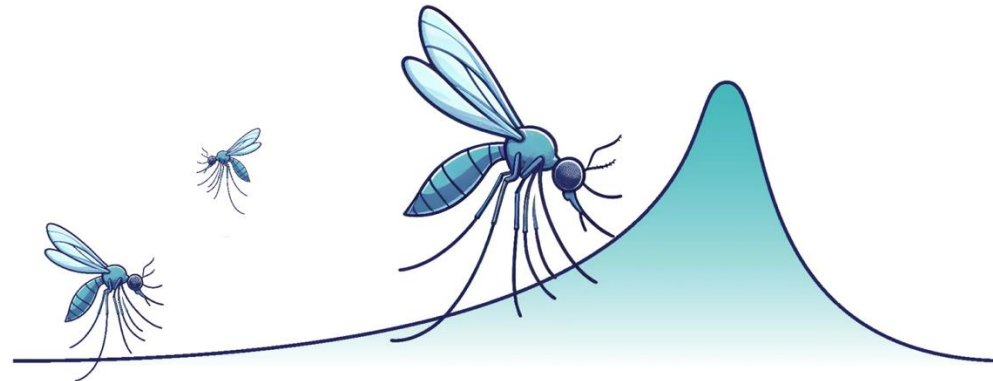


From climate sensitivity to interventions: Work at Heidelberg Planetary Health Hub on *Aedes Aegypti* and Dengue

Julian Heidecke

Climate-Sensitive Vector Dynamics Modelling, Workshop, 18.09.2025



Content

Thermal biology & trait-based models

- Fitting trait-performance curves in a Bayesian hierarchical framework
- Lancet Countdown indicators

Statistical models: Inference, predictions, and Interventions

- Local weather, El Niño & interventions in Sri Lanka
- Teleconnections for Dengue EWS

Thermal biology experiments and trait-based models

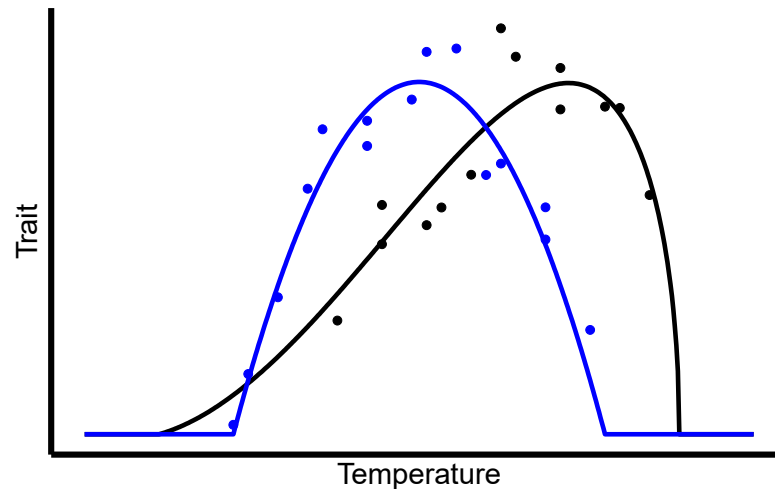
OPEN ACCESS Freely available online



Vectorial Capacity of *Aedes aegypti*: Effects of Temperature and Implications for Global Dengue Epidemic Potential

Jing Liu-Helmersson^{1*}, Hans Stenlund¹, Annelies Wilder-Smith^{1,2}, Joacim Rocklöv¹

¹ Department of Public Health and Clinical Medicine, Epidemiology and Global Health, Umeå University, Umeå, Sweden, ² Lee Kong Chian School of Medicine, Nanyang Technological University, Singapore, Singapore

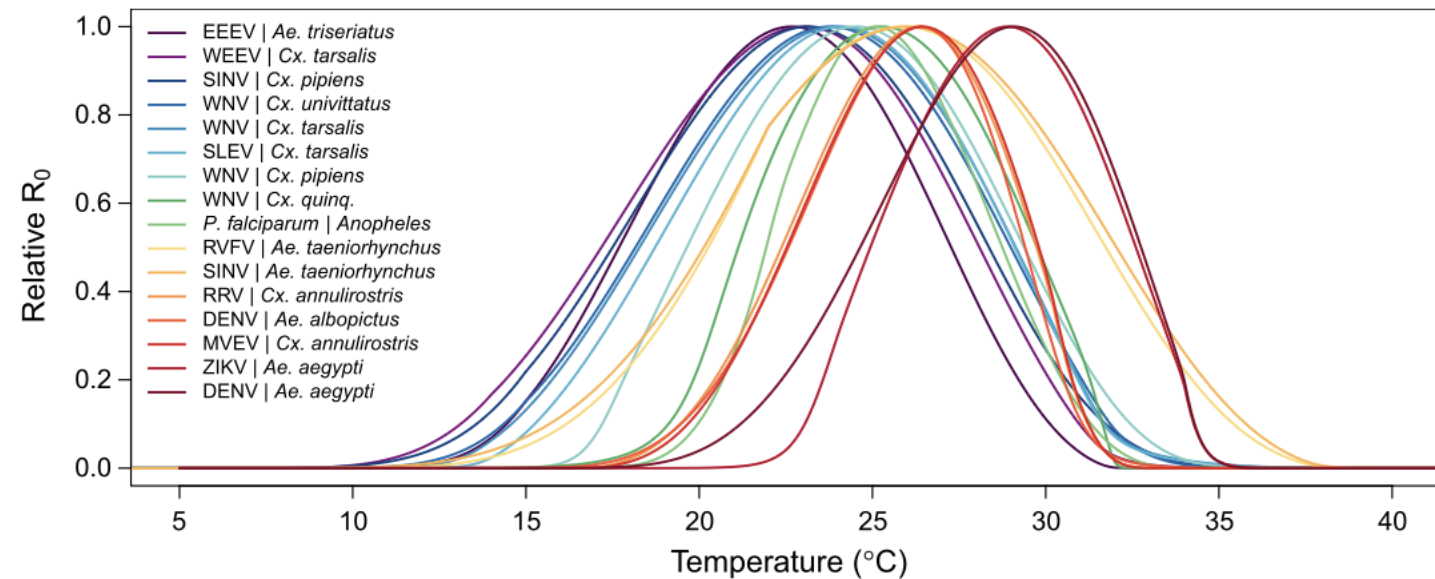


- Quantify effects of temperature on mosquito-pathogen traits
- Parameterize trait-based mathematical models
 - General and interpretable models
 - Models should be challenged by observational data

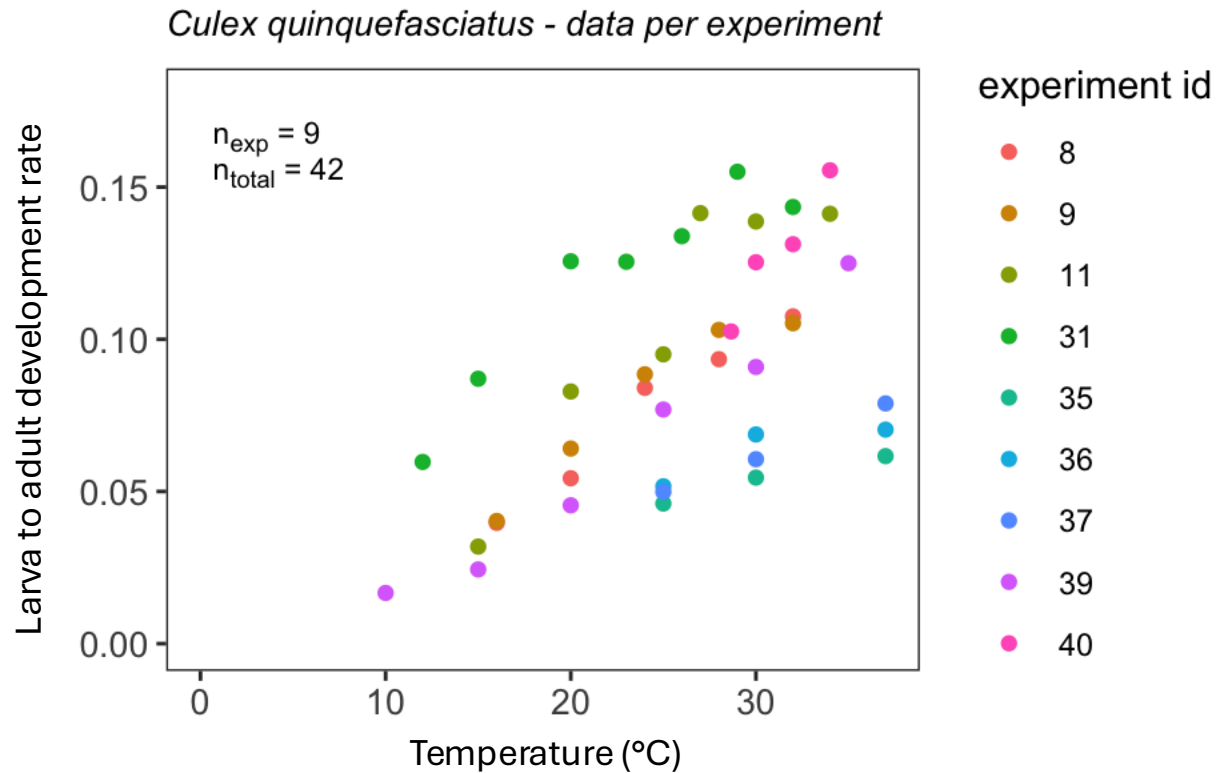
REVIEW AND SYNTHESIS

Thermal biology of mosquito-borne disease

Erin A. Mordecai,^{1*} 
 Jamie M. Caldwell,¹
 Marissa K. Grossman,²
 Catherine A. Lippi,³
 Leah R. Johnson,⁴ Marco Neira,⁵
 Jason R. Rohr,⁶ 
 Sadie J. Ryan,^{3,7} Van Savage,^{8,9}
 Marta S. Shocket,¹ 
 Rachel Sippy,^{3,10}
 Anna M. Stewart Ibarra,¹⁰
 Matthew B. Thomas² and
 Oswaldo Villena⁴



Between-experiment variability

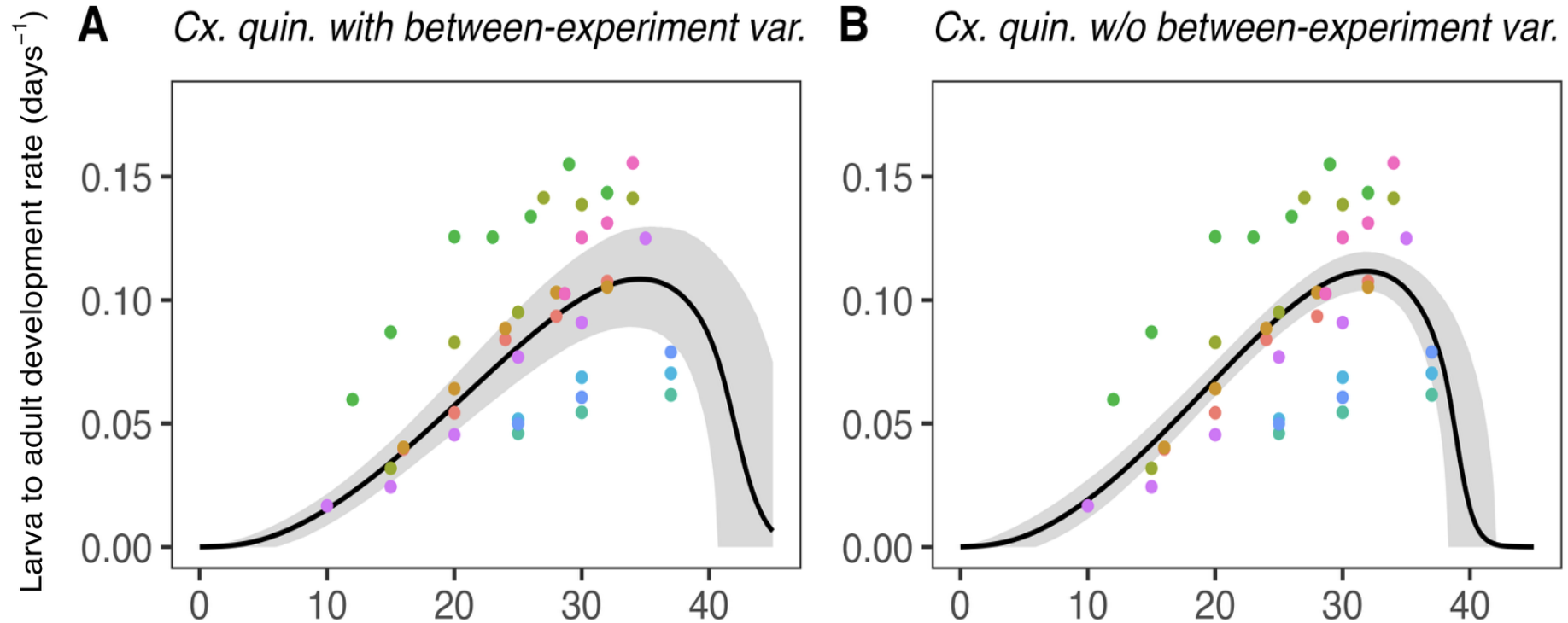


The distribution generating these data points depends on the experiment ID

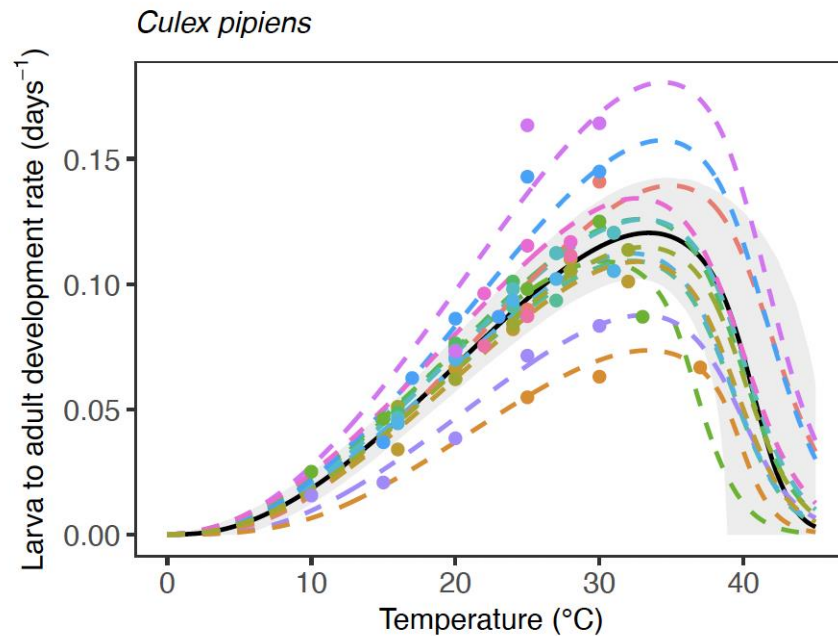
→ Experimental setup?

→ Mosquito population of different origin

Between-experiment variability – Does it matter?



Bayesian hierarchical models



Measurement-level

$$y_{ij,T} | \theta_{ij}, s \sim N(f(T; \theta_{ij}), s^2)$$

Experiment-level

$$\theta_{ij} | \theta_i, \sigma^{\text{exp}} \sim N(\theta_i, \text{diag}(\sigma^{\text{exp}2}))$$

Species-level

$$\theta_i | \mu, \sigma \sim N(\mu, \text{diag}(\sigma^2))$$

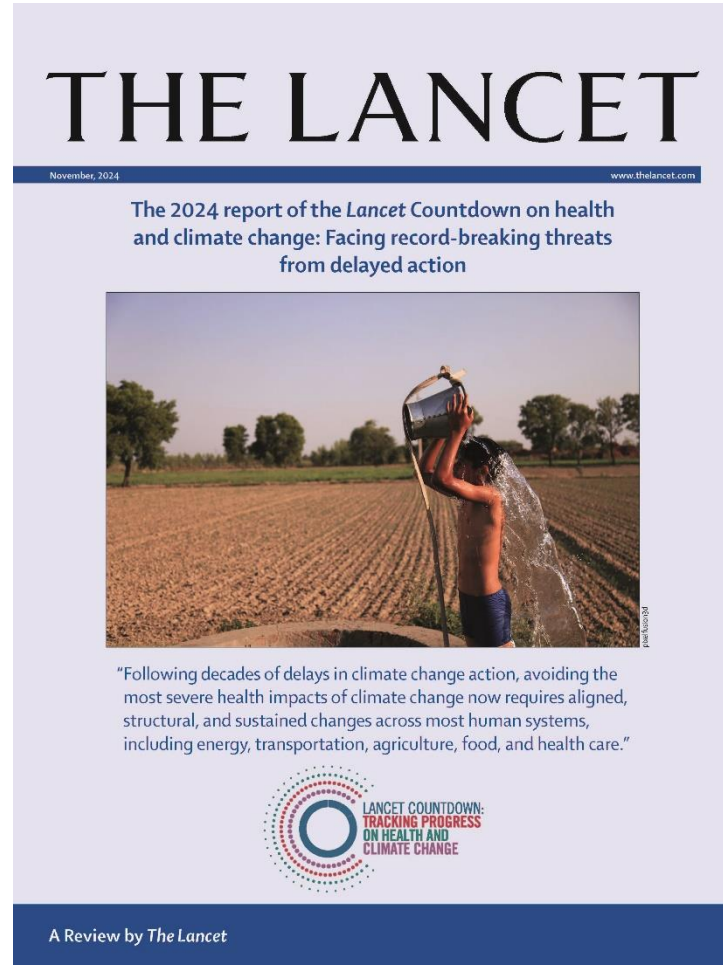
Population-level

μ : mean of parameters across species

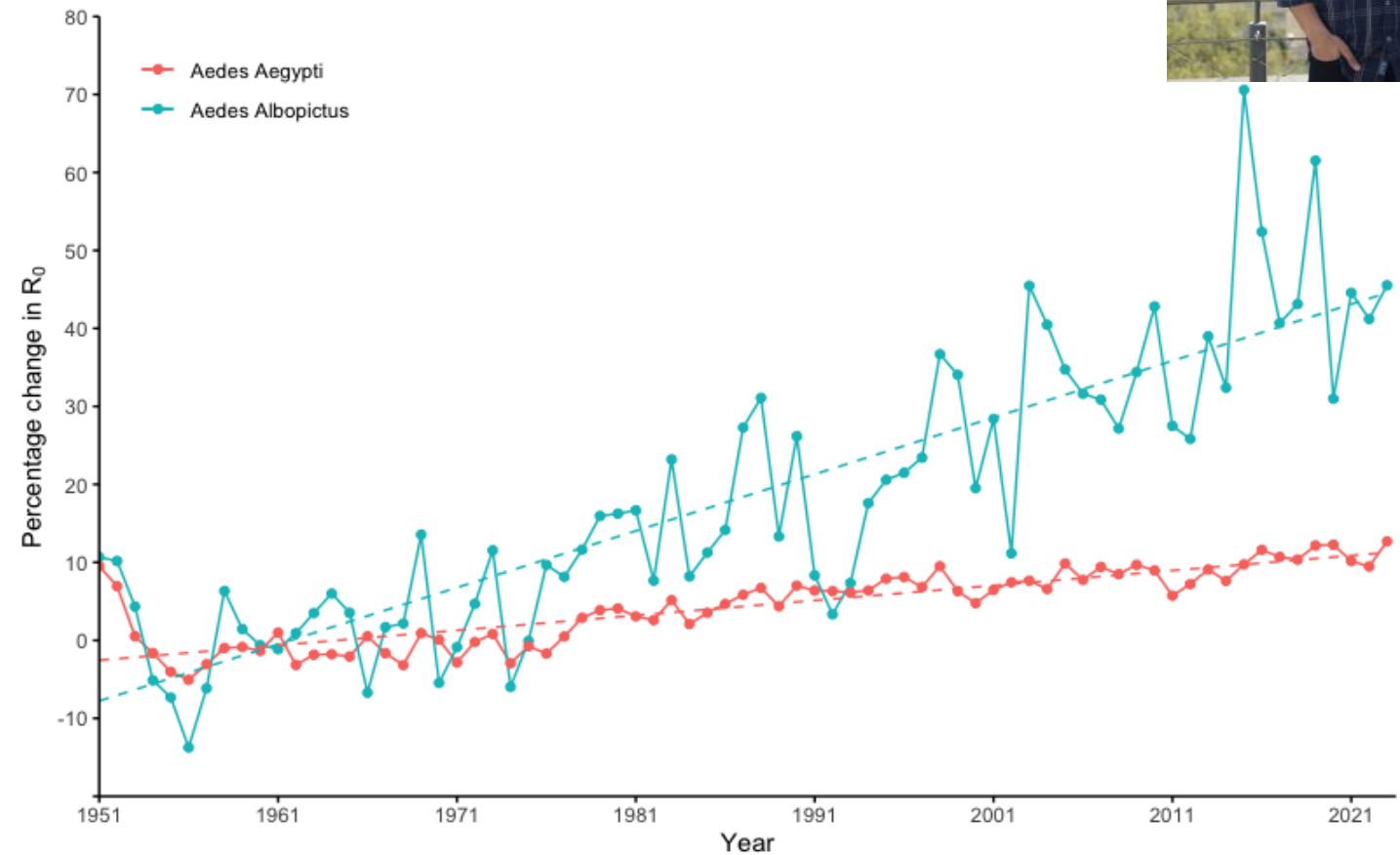
σ : between-species variability

σ^{exp} : between-experiment variability

Lancet Countdown Indicators



Globally aggregated change in Dengue R_0



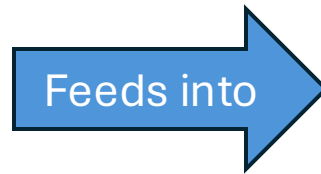
Mathematical model for Dengue-Aegypti

Stage structured population model

$$\frac{dL}{dt} = f\phi h q Z_e(L) M - (\sigma_l + \mu_l) L$$

$$\frac{dP}{dt} = \sigma_l L - (\sigma_p + \mu_p) P$$

$$\frac{dM}{dt} = \sigma_p P - \mu_M M$$



Ross-Macdonald type reproduction number

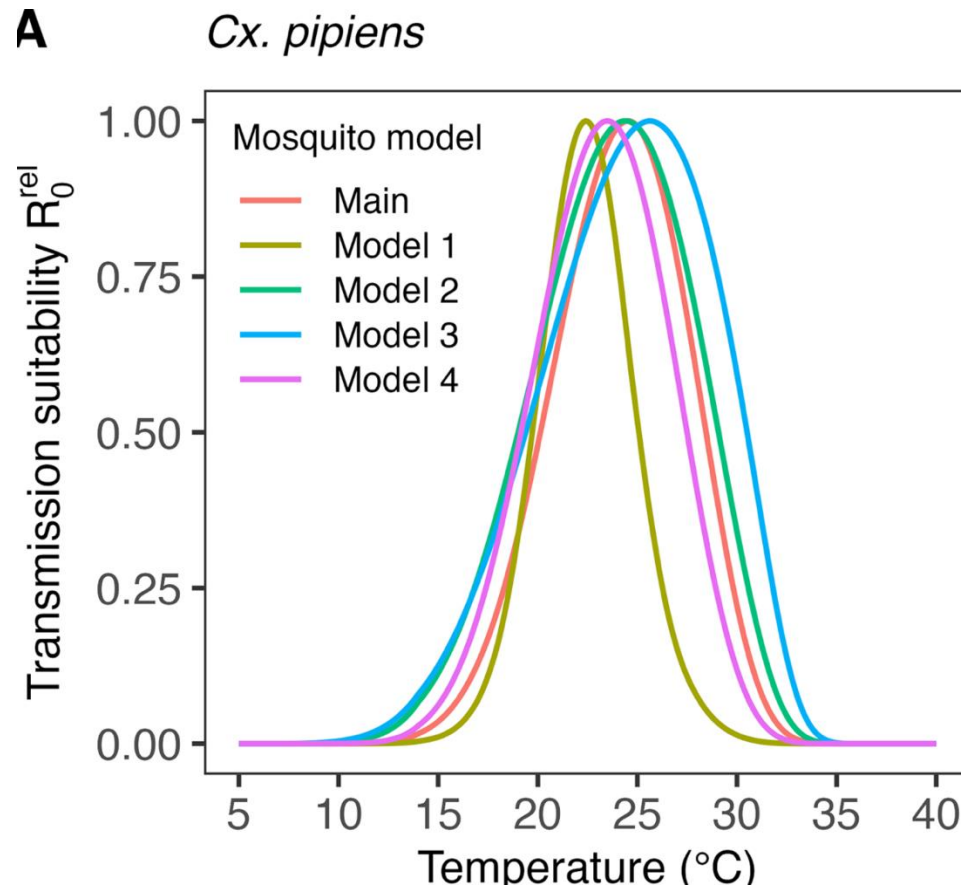
$$R_0 = \frac{\frac{M}{H} a^2 b_H b_M e^{-\mu_M n}}{\gamma_H \mu_H \mu_M}$$

 Temperature

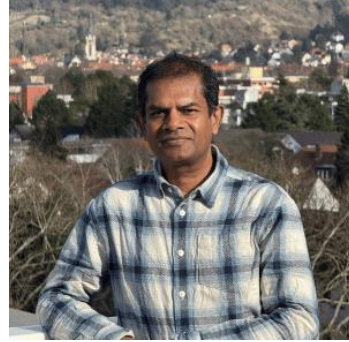
 Human population

 Rainfall

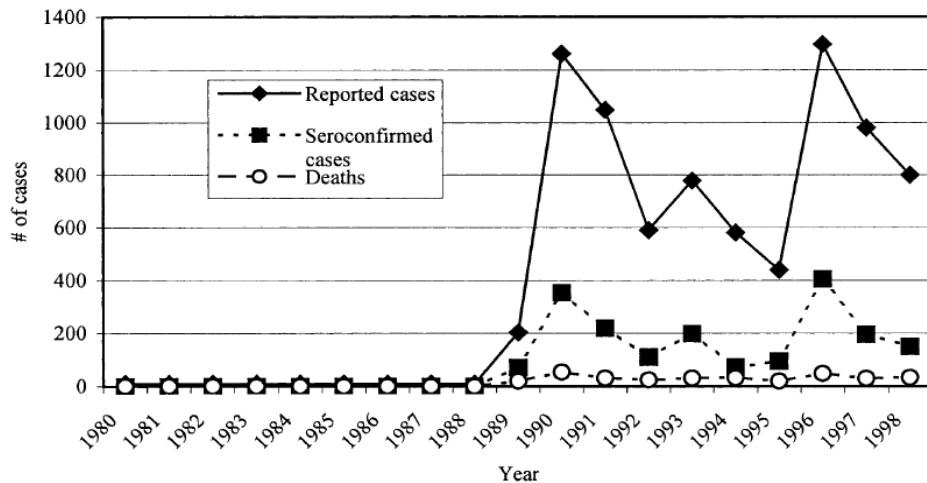
Model structure uncertainty



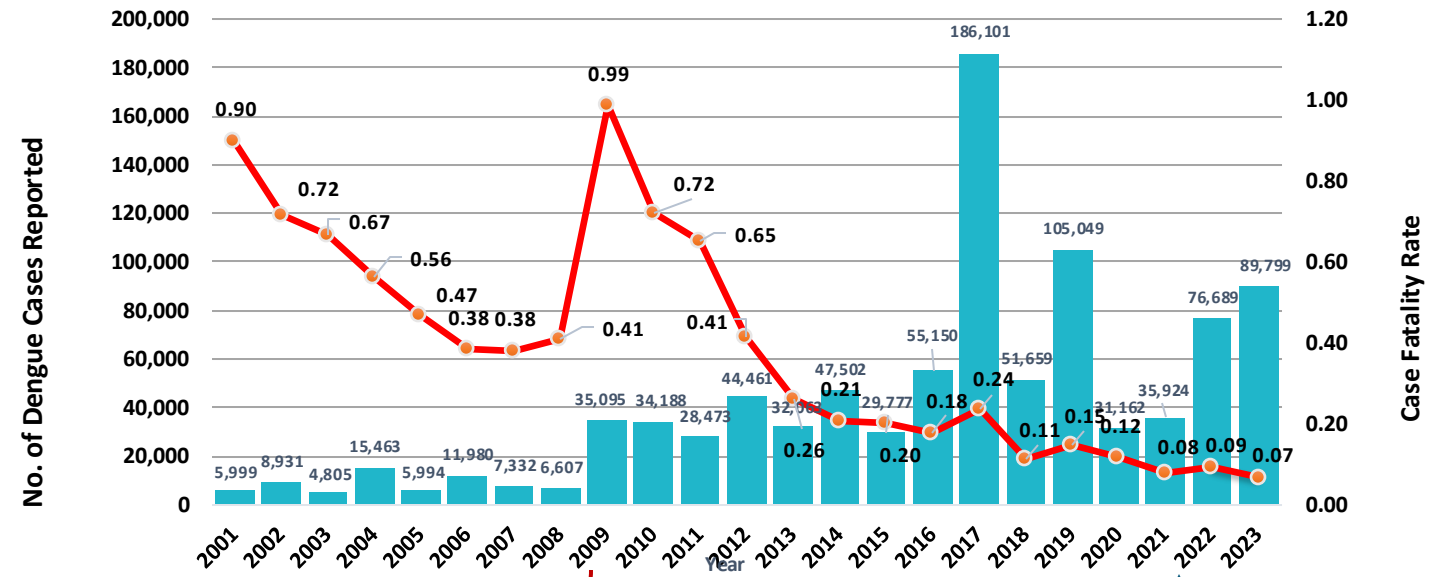
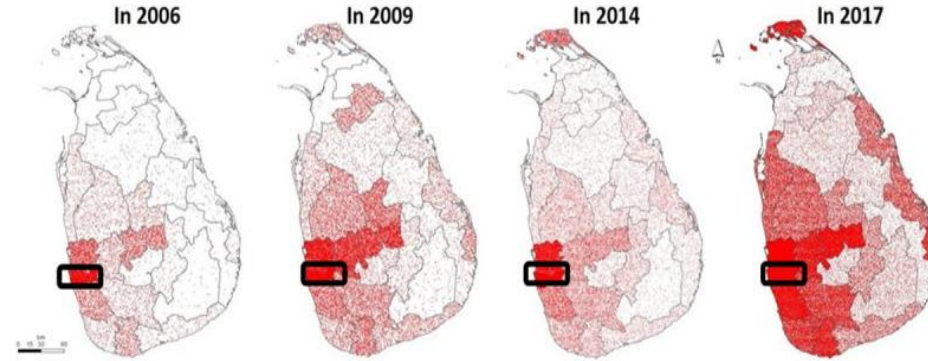
Dengue in Sri Lanka



DHF cases in Sri Lanka (Source: MRI and Epidemiology Unit, Colombo)



More than 20 years before a surge in DHF in Colombo

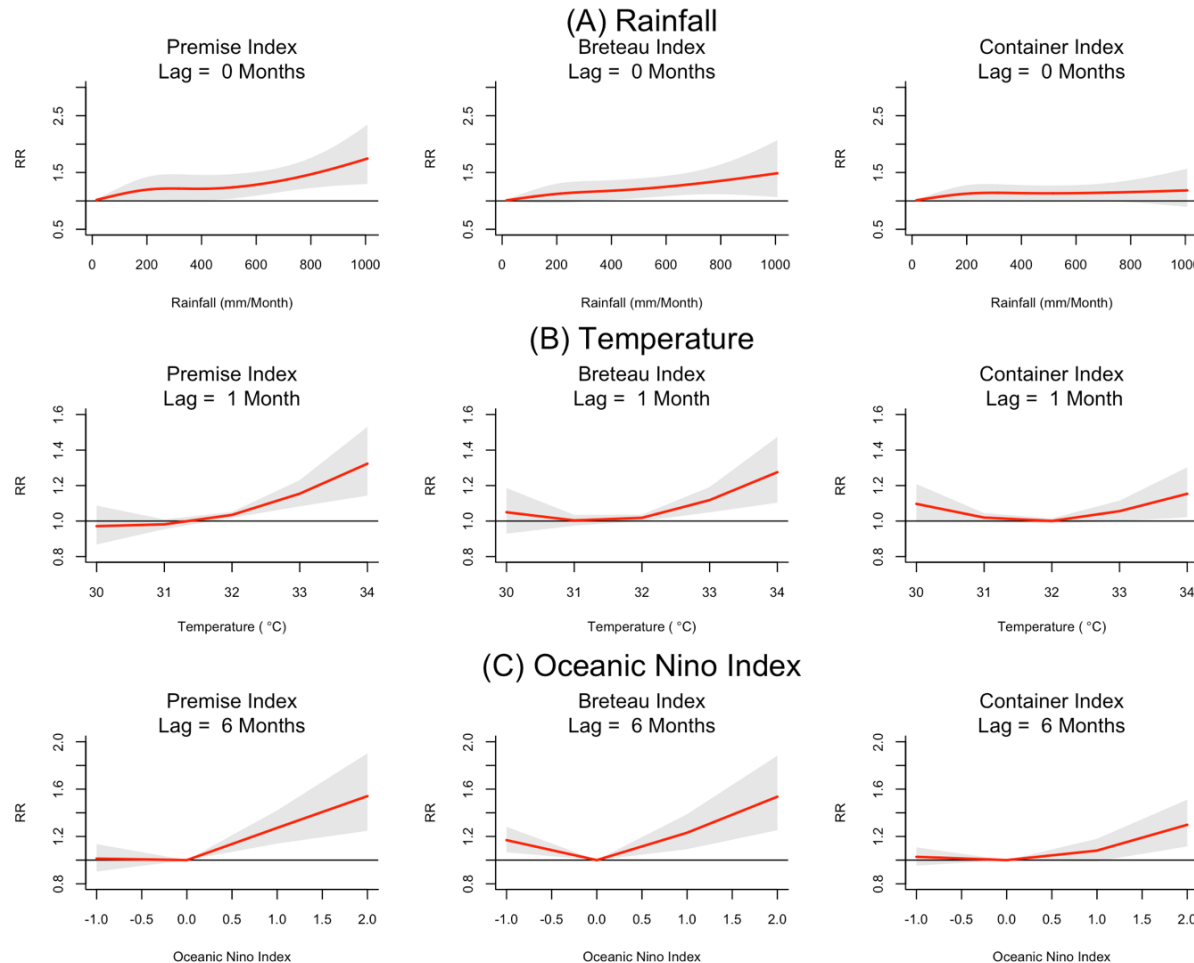


Another 20 years to new surge as the virus spreads across the country

(initial epidemics in the 1990s and early 2000 were confined mostly to the Western Province and the Colombo district)

Colombo Force of infection: 14.9
Kandy District: 1.3

Effect of El Niño–Southern Oscillation (ENSO) and weather on *Aedes* vectors in Kalutara, Sri Lanka, 2010-2018



Rainfall:

- Lag 0 months
- >400mm/month (for PI and BI)

Temperature:

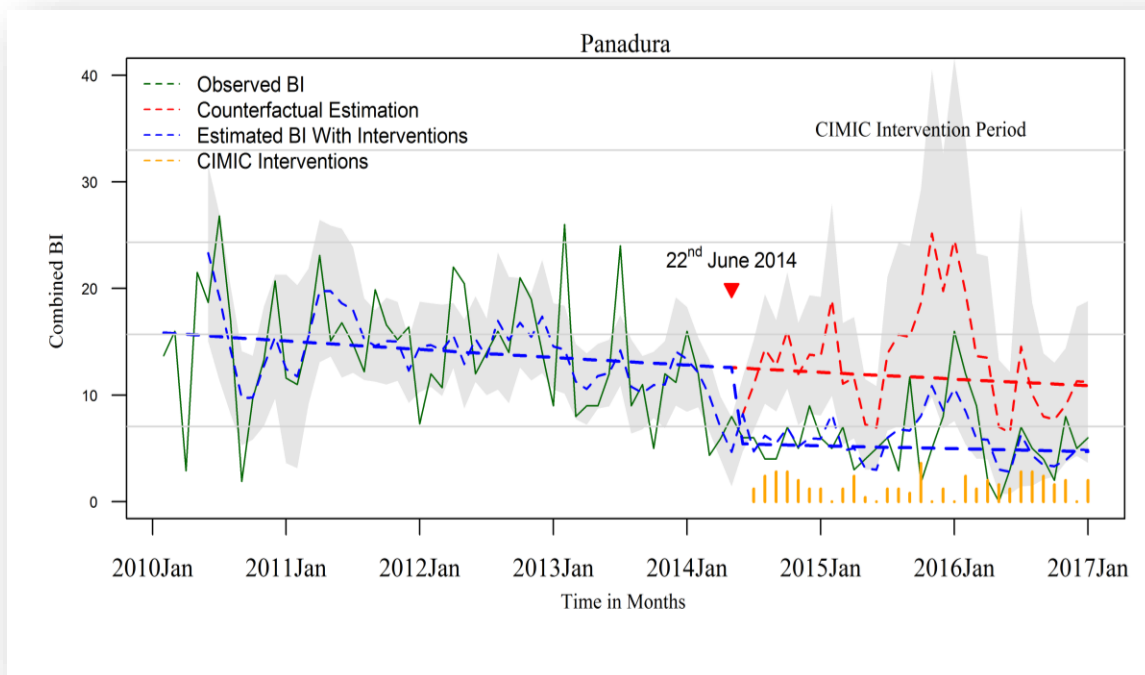
- Lag 1 to 2 months
- >31.5°C

Oceanic Nino index (ONI):

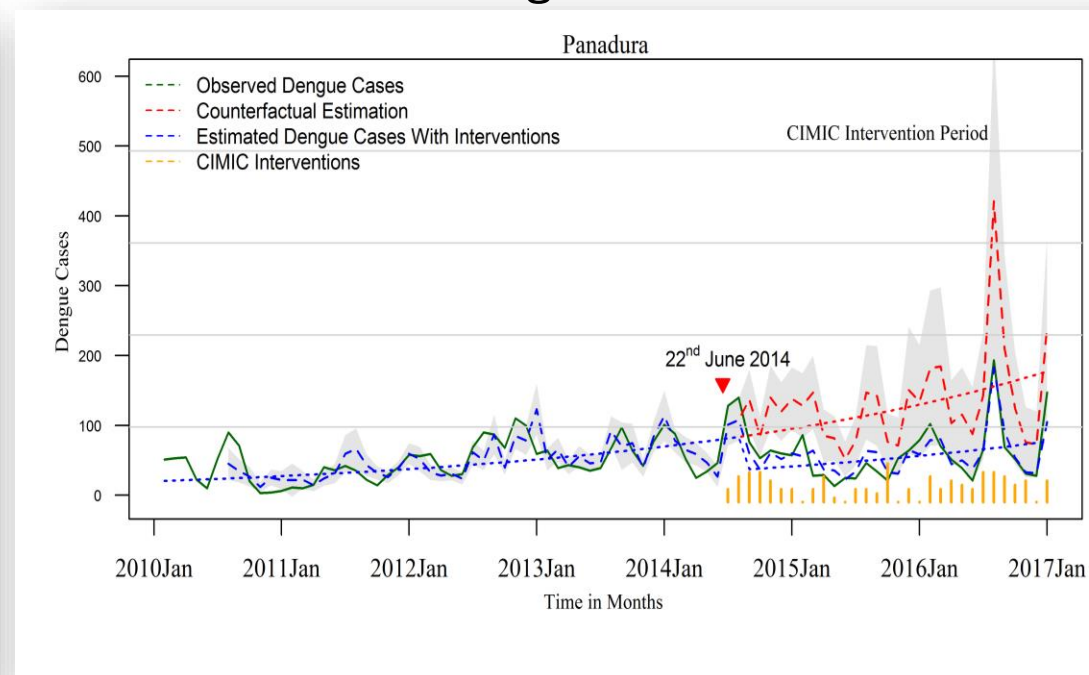
- 6 months
- ONI > 0 (El Niño) for PI, BI and CI

Evaluation of intensified dengue control measures in Pandura, Sri Lanka

Breteau Index

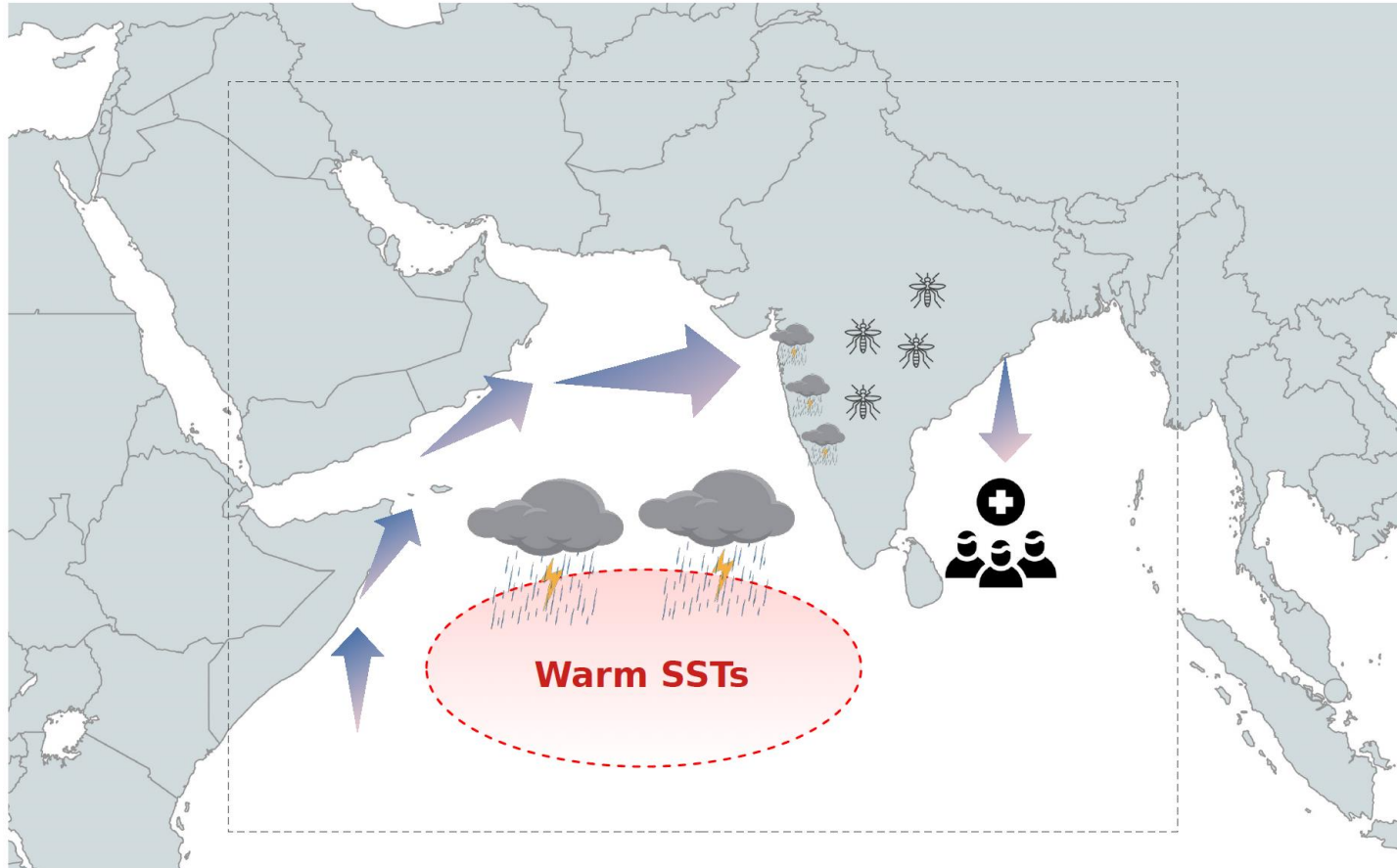


Dengue Cases

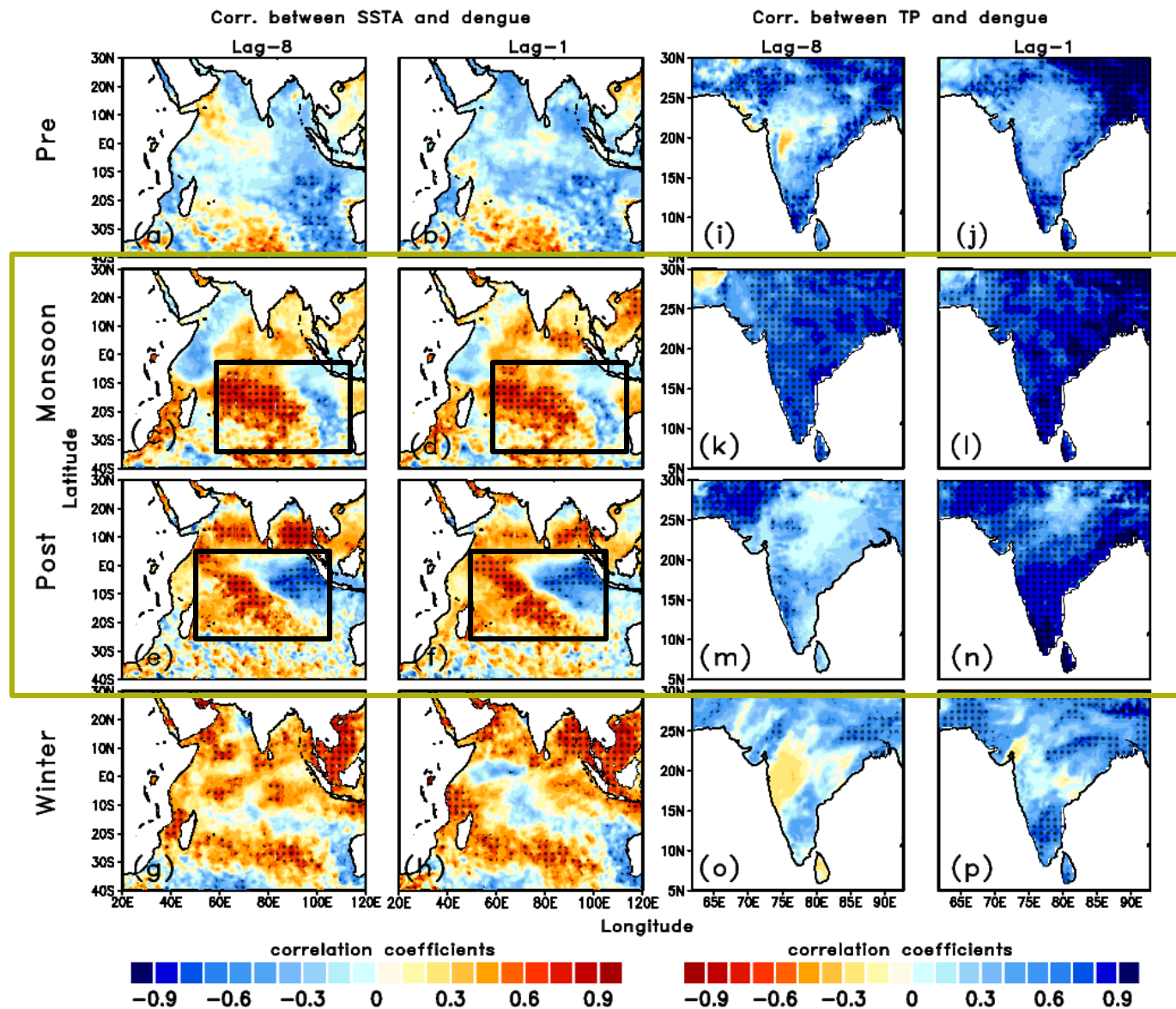


Estimated mean number of averted dengue cases: 2192 (95% CI: 1741-2643)

Response of Dengue to sea surface temperature structures



- Warm SSTs lead to increased evaporation
- Winds transport moisture toward continental land
- Modulates rainfall variability
- Drives mosquito population dynamics
- Affects Dengue risk



- In South Asia → dengue case counts are stat. Sign. correlated with SSTs in the western Indian Ocean during the monsoon and post-monsoon season
- travel-related dengue cases appear to be associated with the positive Indian Ocean Dipole pattern and SST anomalies over the subtropical southern Indian Ocean.

Thank you!



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